Unit 06: Extracting Data Sets

CONTENTS

Objectives

After this lecture, you will be able to

- Know the process of accessing and loading information from the Repository into the Process using retrieve Operator.
- Implementation of storage operator to store data and model.
- Various methods of visualizing the data.
- Creation of a new repository and usage of an existing repository.

Introduction

Following the directed dialogue or using the drag and drop feature to import data to your repository is easy. Simply drag the file from your file browser onto the canvas and follow the on-screen instructions. Check that the data types are right and that the goal or mark is correctly flagged. The fact that this "import then open" method is not like other methods of data opening is a significant difference.

6.1 Read Excel

Read excel operator reads an ExampleSet from the specified Excel file.

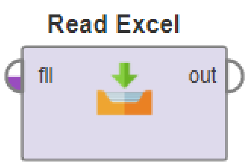


Figure 1: Read Excel operator

Data from Microsoft Excel spreadsheets can be loaded using this operator. Excel 95, 97, 2000, XP, and 2003 data can be read by this operator. The user must specify which of the workbook's spreadsheets will be used as a data table. Each row must represent an example, and each column must represent an attribute in the table.Please keep in mind that the first row of the Excel sheet can be used for attribute names that are defined by a parameter. The data table can be put anywhere on the sheet and can include any formatting instructions, as well as empty rows and columns. Empty cells or cells with only "?" can be used to show missing data values in Excel.

The Design perspective is your creative canvas and the location where you will spend the majority of your time. You will merge operators into data mining processes here.

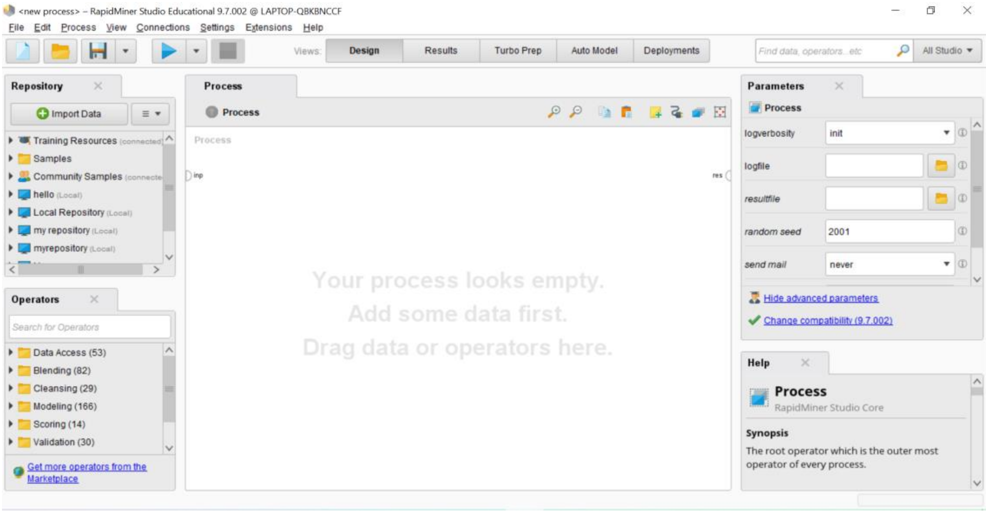


Figure 2: Design view

In the Repository view, tomake a new folder:

1. Right-click on Local Repository and select Properties.

2. Select Create Folder from the drop-down menu.

3. Give the new folder a name, such as Getting Started, and then click OK.

Connect a data folder and a processes folder to the procedure from the previous step.Your Repositories view should look similar to this:

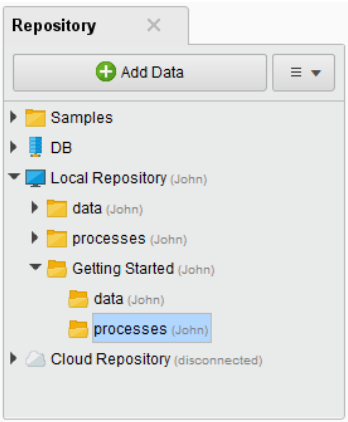


Figure 3: Repository View

The easiest and shortest way to import an Excel file is to use the *import configuration wizard* from the Parameters panel. The easiest approach, which could take a little more time, is to set all of the parameters in the Parameters panel first, then use the wizard. Before creating a method that uses the Excel file, please double-check that it can be read correctly.

To get started with RapidMiner Studio, build a local repository on your computer.

From the Repositories view, select Import Excel Sheet from the pull-down to import the training data set.

The Import Wizard launches. Browse to the location where you saved customer-churn-data.xlsx, select the file, and click Next.

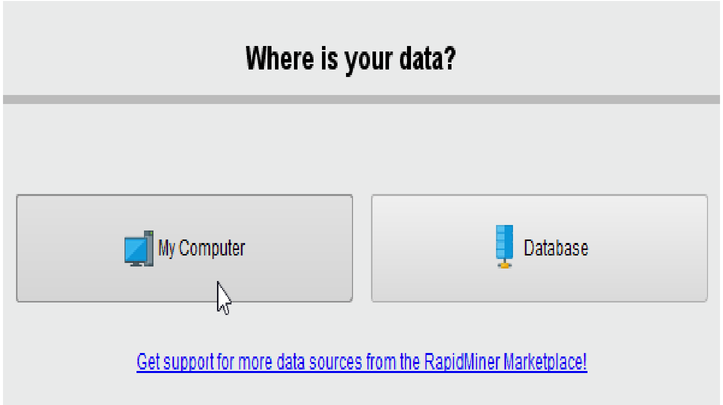
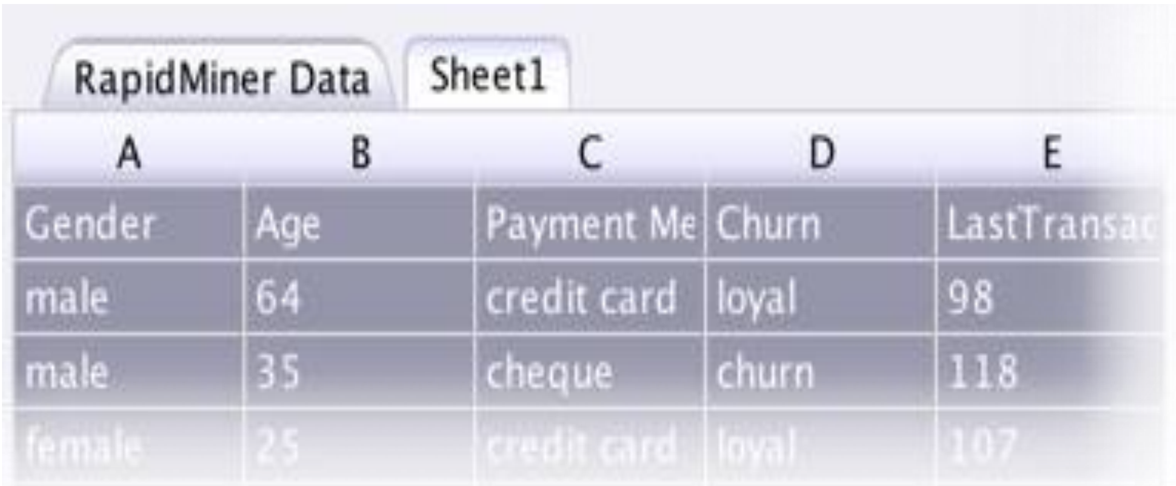
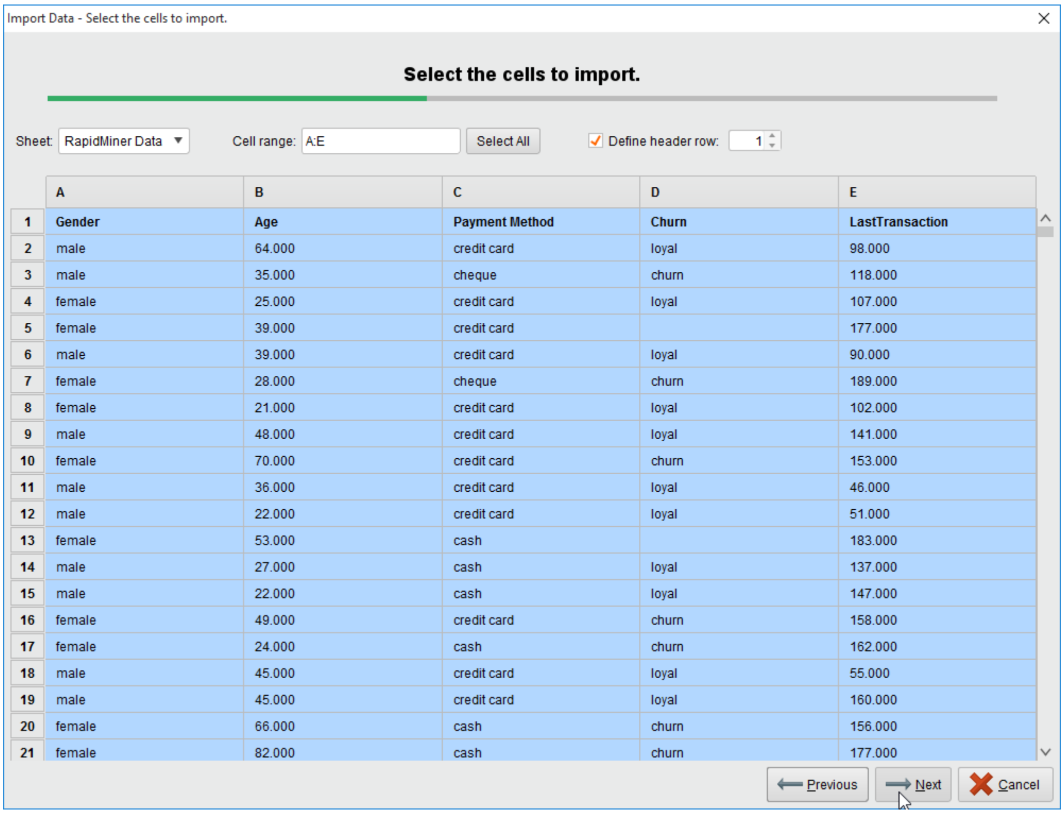


Figure 4:Data Source Selection

The wizard will walk you through the process of importing data into the repository. Verify that you're importing the right Excel sheet by looking at the tabs at the end. Although there is only one sheet in this file, RapidMiner Data, it is always a good idea to double-check.If there were more sheets, it may look like this:



Step 2 also allows you to select a range of cells for import. For this tutorial, you want all cells (the default). Click Next.

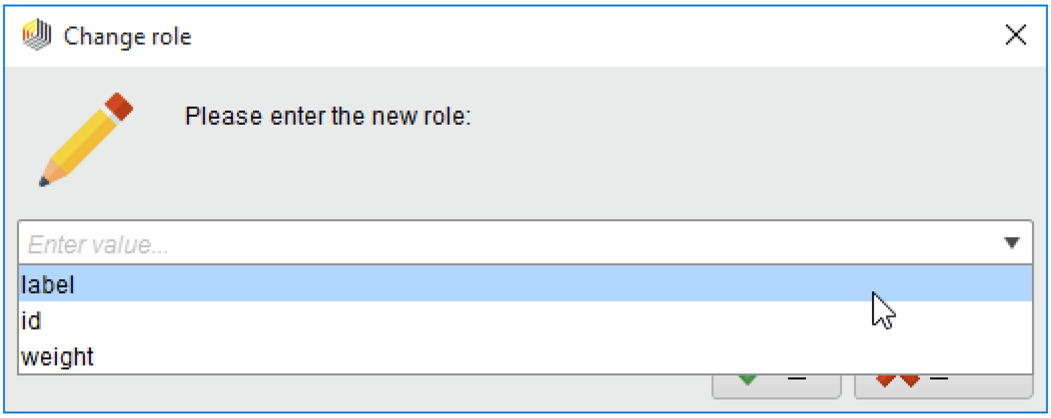


	A	B	C	D	E
1	Gender	Age	Payment Method	Churn	LastTransaction
2	male	64.000	credit card	loyal	98.000
3	male	35.000	cheque	churn	118.000
4	female	25.000	credit card	loyal	107.000
5	female	39.000	credit card		177.000
6	male	39.000	credit card	loyal	90.000
7	female	28.000	cheque	churn	189.000
8	female	21.000	credit card	loyal	102.000
9	male	48.000	credit card	loyal	141.000
10	female	70.000	credit card	churn	153.000
11	male	36.000	credit card	loyal	46.000
12	male	22.000	credit card	loyal	51.000
13	female	53.000	cash		183.000
14	male	27.000	cash	loyal	137.000
15	male	22.000	cash	loyal	147.000
16	female	49.000	credit card	churn	158.000
17	female	24.000	cash	churn	162.000
18	male	45.000	credit card	loyal	55.000
19	male	45.000	credit card	loyal	160.000
20	female	66.000	cash	churn	156.000
21	female	82.000	cash	churn	177.000

RapidMiner has preselected the first row as the row that contains column names in this process. You could fix it here if it was inaccurate, but this isn't important for your data collection. Accept the default and move on to the next step. Define the data that will be imported. The example set consists of the entire spreadsheet, with each row or example representing one customer.

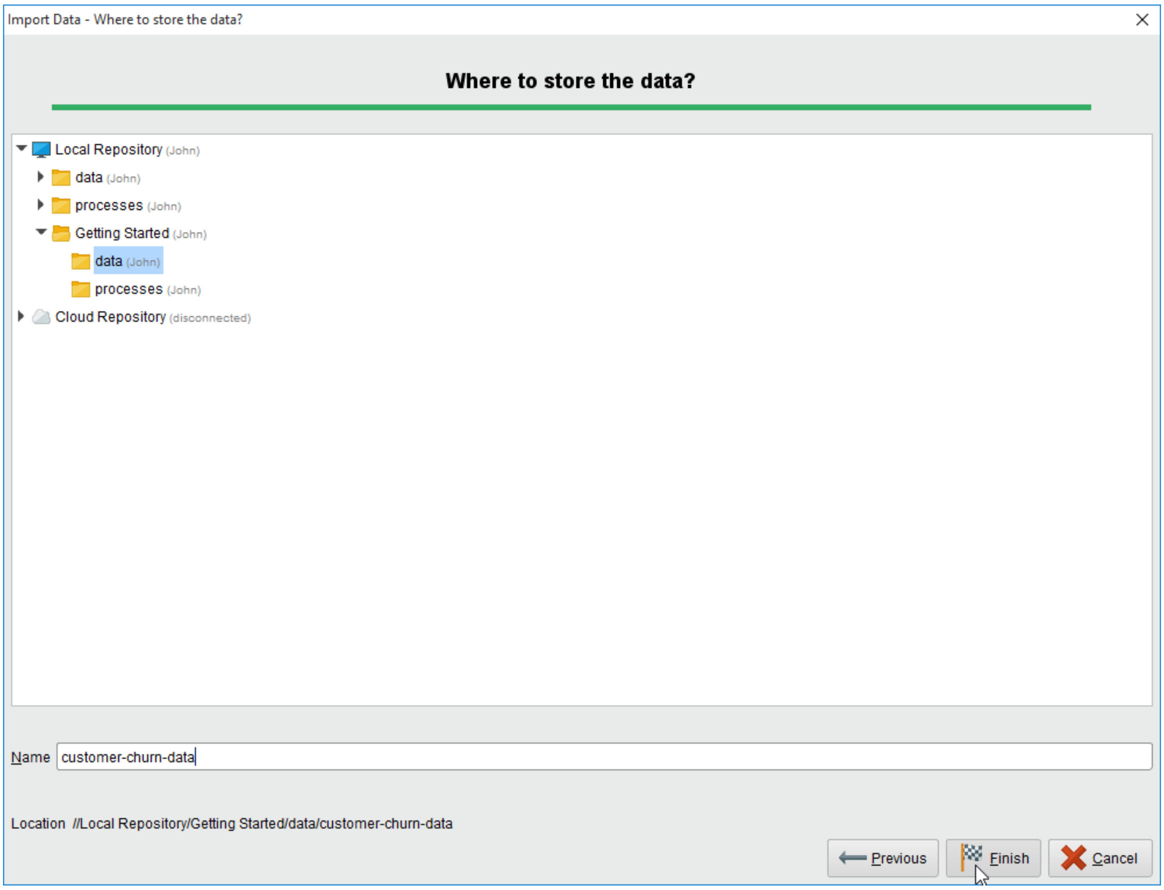
This phase has four essential components: The specifies the columns should be imported. The column names (or attributes, as RapidMiner refers to them) are those that were defined in the previous phase by the Name annotation. These are Gender, Age, Payment Method, Churn, and LastTransaction.The data types for each attribute are described by the drop-down boxes in the third row. The data type determines the values are permitted for an attribute (polynomial, numeric, integer, etc.).

The pull-down showing attribute in the fourth row is the most important task for this import. This pull-down allows you to tell RapidMiner which attribute is the main focus of your model. You set the function of Churn to mark because it is the column you want to forecast. RapidMiner can identify the example by predicting a value for each label based on what it has learned.



To finish the import, navigate to the data folder in Getting Started and give the file a name. It's worth noting that while it's in the RapidMiner archive, it'll be saved in RapidMiner's special format, which means it won't have a file name extension by convention. Click on Finish.

When you import data in RapidMiner, you need to select the attribute type "label" for the column you wish to classify.



When you click Finish, the data set loads into your repository and RapidMiner switches to the Resultsperspective, where your data displays. The following parameters you need to set to import your data successfully.

Person research tasks should be organized into new folders in the repository, which should be named appropriately.

Input

An Excel file is expected to be a file object that can be generated by using other operators that have file output ports, such as the Read File operator.

Output

This port provides a tabular version of the Excel file as well as metadata. This contribution is comparable to that of the Retrieve operator.

Parameters

- **import_configuration_wizard:** This choice allows you to use a wizard to configure this operator. This operator is simple to use thanks to this user-friendly wizard.
- **excel_file:**The path of the Excel file is specified here. It can be selected using the choose a file button.

- **sheet_selection:** This option allows you to change the sheet selection between sheet number and sheet name.
- **sheet_number:** The number of the sheet which you want to import should be specified here.Range: integer
- **sheet_name:** The name of the sheet which you want to import should be specified here.Range: string
- **imported_cell_range:** This is a mandatory parameter. The range of cells to be imported from the specified sheet is given here. It is specified in 'xm:yn' format where 'x' is the column of the first cell of the range, 'm' is the row of the first cell of the range, 'y' is the column of the last cell of the range, 'n' is the row of the last cell of the range. 'A1:E10' will select all cells of the first five columns from rows 1 to 10.
- **first_row_as_names:** If this option is set to true, it is assumed that the first line of the Excel file has the names of attributes. Then the attributes are automatically named and the first line of the Excel file is not treated as a data line.Range: boolean
- **annotations:** If the first row as names parameter is not set to true, annotations can be added using the 'Edit List' button of this parameter which opens a new menu. This menu allows you to select any row and assign an annotation to it. Name, Comment, and Unit annotations can be assigned. If row 0 is assigned Name annotation, it is equivalent to setting the first row as names parameter to true. If you want to ignore any rows you can annotate them as Comments.
- **date_format:** The date and time format are specified here. Many predefined options exist; users can also specify a new format. If text in an Excel file column matches this date format, that column is automatically converted to date type. Some corrections are automatically made in the date type values. For example, a value '32-March' will automatically be converted to '1-April'. Columns containing values that can't be interpreted as numbers will be interpreted as nominal, as long as they don't match the date and time pattern of the date format parameter. If they do, this column of the Excel file will be automatically parsed as the date and the according attribute will be of date type.
- **time_zone:** This is an expert parameter. A long list of time zones is provided; users can select any of them.
- **Locale:** This is an expert parameter. A long list of locales is provided; users can select any of them.
- **read_all_values_as_polynominal:** This option allows you to disable the type handling for this operator. Every column will be read as a polynomial attribute. To parse an excel date afterwards use 'date_parse(86400000 * (parse(date_attribute) - 25569))' (- 24107 for Mac Excel 2007) in the Generate Attributes operator.Range: boolean

- **data_set_meta_data_information:** This option is an important one. It allows you to adjust the metadata of the ExampleSet created from the specified Excel file. Column index, name, type, and role can be specified here. The Read Excel operator tries to determine an appropriate type of attribute by reading the first few lines and checking the occurring values. If all values are integers, the attribute will become an integer. Similarly, if all values are real numbers, the attribute will become of type real. Columns containing values that can't be interpreted as numbers will be interpreted as nominal, as long as they don't match the date and time pattern of the date format parameter. If they do, this column of the Excel file will be automatically parsed as the date and the according attribute will be of type date. Automatically determined types can be overridden using this parameter.
- **read_not_matching_values_as_missings:** If this value is set to true, values that do not match with the expected value type are considered as missing values and are replaced by '?'. For example, if 'back is written in an integer column, it will be treated as a missing value. A question mark (?) or an empty cell in the Excel file is also read as a missing value.Range: boolean

6.2 Store Operator

Store operator a place in the data repository where an IO Object is stored The repository entry parameter specifies the location of the object to be stored. Using the Retrieve operator, other processes may access the stored object. Please see the attached Example Processes for a basic understanding of how this operator works. In the Example, the Store operator is used to store an ExampleSet and a model.

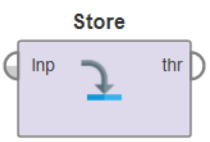


Figure 5:Store Operator

Storing an ExampleSet using the Store operator

The following procedure demonstrates how to store an ExampleSet using the Store operator. The Retrieve operator is used to load the 'Golf' and 'Golf-Testset' data sets. The Append operator is used to combine these ExampleSets. The name of the resulting ExampleSet is 'Golf-Complete,' and it is saved using the Store operator. In the third Example Method, the stored ExampleSet is used.

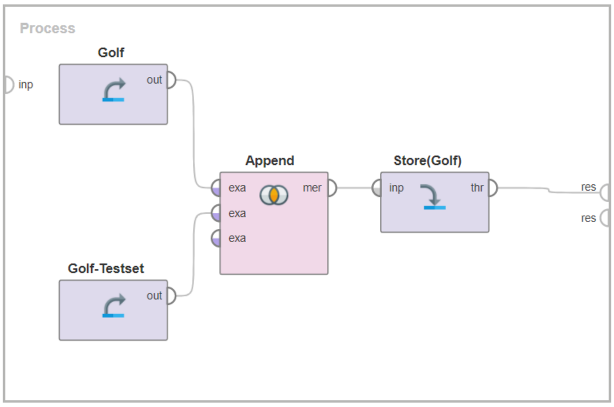


Figure 6: Storing Golf dataset using Store Operator

Create two datasets and merge the results of both datasets and store the merged results using store operator.

Storing a model using the Store operator

The following procedure demonstrates how to use the Store operator to save a model. The Retrieve operator is used to load the 'Golf' data set. It is subjected to the Naive Bayes operator, and the resulting model is saved in the repository using the Store operator. 'Golf-Naive-Style' is the name given to the model. In the third Example Method, the stored model is used.

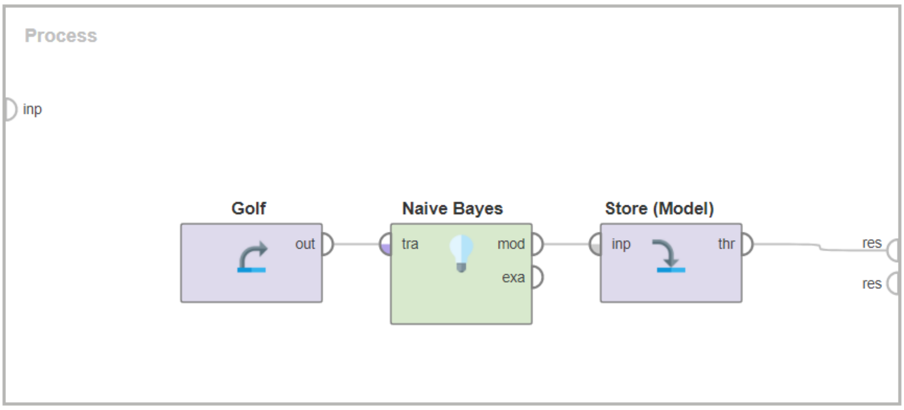


Figure 7: Storage of Naive Bayes using Store Operator

There are two quick and easy ways to store a RapidMiner model in a repository

1. Right-click on the tab in the **Results panel** and you should see an option to store the model in the repository:

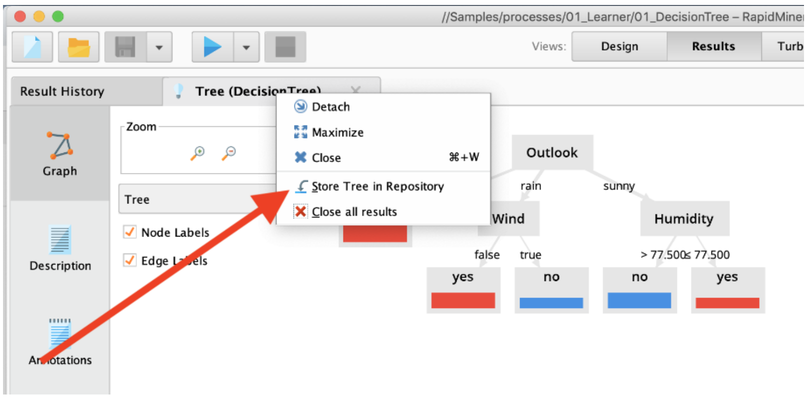


Figure 8: Storage using result panel

2.Use the Store operator on any green model "wire".

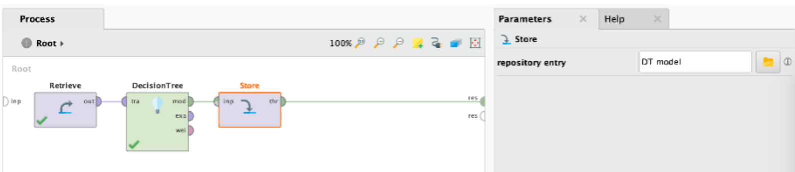


Figure 9: Store operator using Design View

Either way will get you a model object in the repository which you can use/view any time you like:

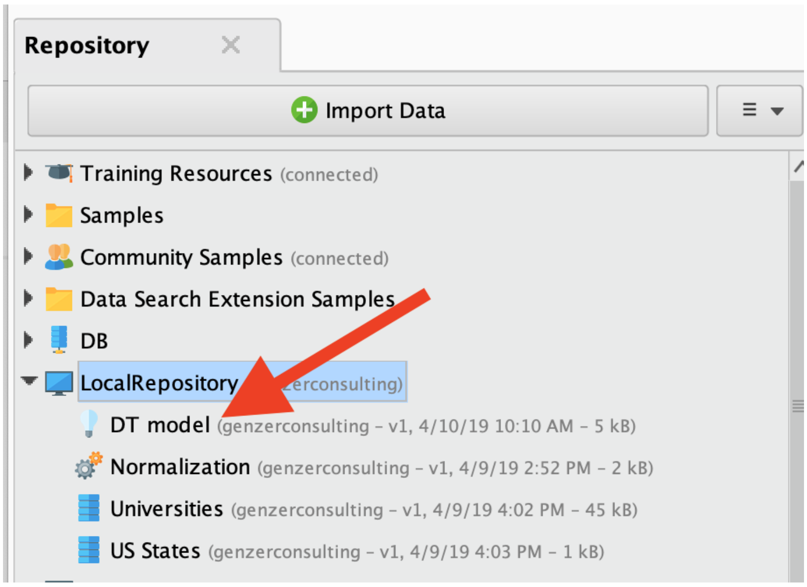


Figure 10: Storage in Repository

6.3 Retrieve Operator

The Retrieve operator a RapidMiner Object is loaded into the Process. This Object is usually an ExampleSet, but it may also be a Collection or a Model. This method of data retrieval also returns the RapidMiner Object's metadata.

Figure 11:Retrive Operator

This Operator is similar to the Data Access group's various Read source> Operators. The benefit of storing data in a repository is that metadata properties are also preserved. Additional information about the RapidMiner Object you retrieve is provided by metadata.

Output

The RapidMiner Object with the path defined in the repository entry parameter is returned.

Parameters

repository_entry

The location of the RapidMiner Object to be loaded. This parameter points to a registry entry, which will be returned as the Operator's output.

Repository locations are resolved relative to the Repository folder containing the current Process. Folders in the Repository are separated by a forward slash ('/'). A '..' references the parent folder. A leading forward slash references the root folder of the Repository containing the current Process. A leading double forward slash ('//') is interpreted as an absolute path starting with the name of a Repository. The list below shows the different methods:

- 'MyData' looks up an entry 'MyData' in the same folder as the current Process
- '../Input/MyData' looks up an entry 'MyData' located in a folder 'Input' next to the folder containing the current Process
- '/data/Model' looks up an entry 'Model' in a top-level folder 'data' in the Repository holding the current Process

- '//Samples/data/Golf' looks up the Iris data set in the 'Samples' Repository.

When using the “Select the repository location” button, it is possible to check if the path should be resolved, relative. This is useful when sharing Processes with others.

Loading of Data using the Retrieve Operator

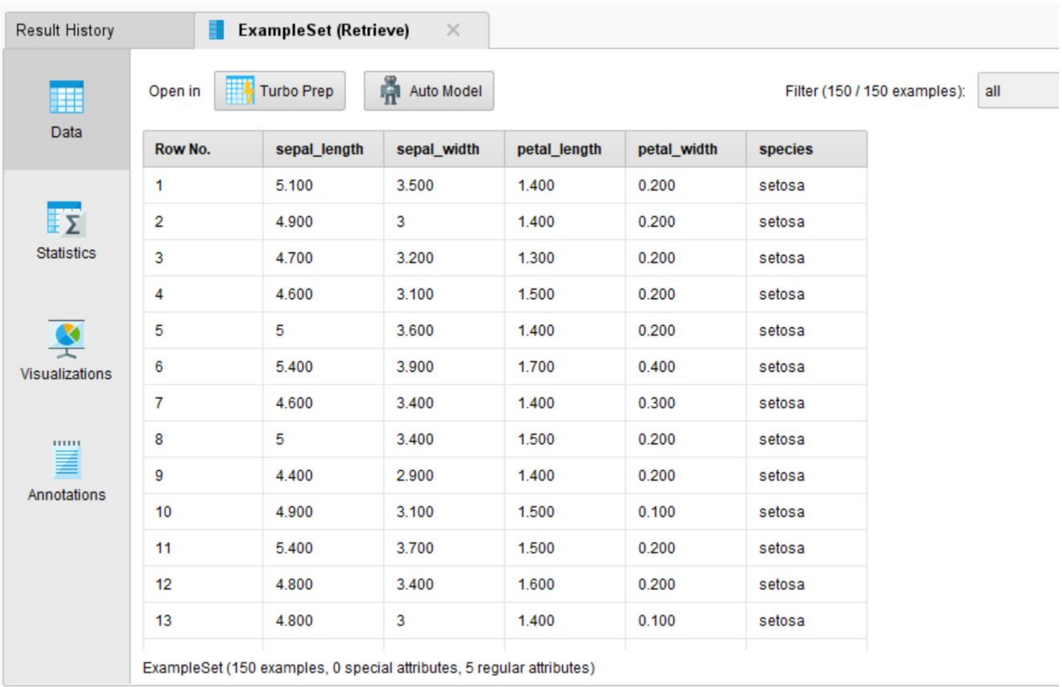
The Golf data set is loaded from the repository using the following procedure. The repository entry parameter is '/Samples/data/Golf', which is an absolute direction. As a result, the Golf data collection and sub-folder data are returned from the Samples repository.

Figure 12:Data Retrieval

Create a student data set and load it in the local repository of rapidminer studio and generate the results.

6.4 Graphical Representation of data in Rapidminer

Objects located at the result ports on the right-hand side of a process are automatically reflected in the Results Perspective after the process is completed, as we've already seen. The wide area on the top left-hand side, where the Results Overview is already shown, is used for this. Iris Dataset is being considered for graphical representation. Figure 13 shows the attributes and corresponding values of the Iris dataset.



Row No.	sepal_length	sepal_width	petal_length	petal_width	species
1	5.100	3.500	1.400	0.200	setosa
2	4.900	3	1.400	0.200	setosa
3	4.700	3.200	1.300	0.200	setosa
4	4.600	3.100	1.500	0.200	setosa
5	5	3.600	1.400	0.200	setosa
6	5.400	3.900	1.700	0.400	setosa
7	4.600	3.400	1.400	0.300	setosa
8	5	3.400	1.500	0.200	setosa
9	4.400	2.900	1.400	0.200	setosa
10	4.900	3.100	1.500	0.100	setosa
11	5.400	3.700	1.500	0.200	setosa
12	4.800	3.400	1.600	0.200	setosa
13	4.800	3	1.400	0.100	setosa

Figure 13:Iris Dataset

2D Scatter Plot

The resulting interface demonstrates a wide range of visualization options; in this case, we'll use RapidMiner's advanced plotting capabilities. Set the domain dimension of the iris dataset to a1, the axis to a2, and the color dimension to the mark in the advanced charts tab. Adding a dimension to the axis of a 2D scatter plot, as shown in Figure 14.

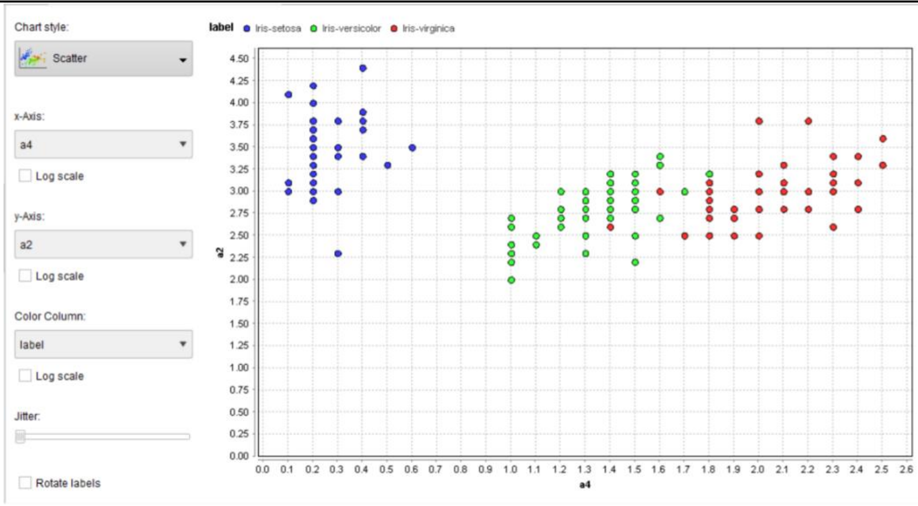


Figure 14:Scatter Plot

Line Chart

If you'd rather make a line chart, the process is the same, but you'll need to change the format to lines as shown in Figure 15.

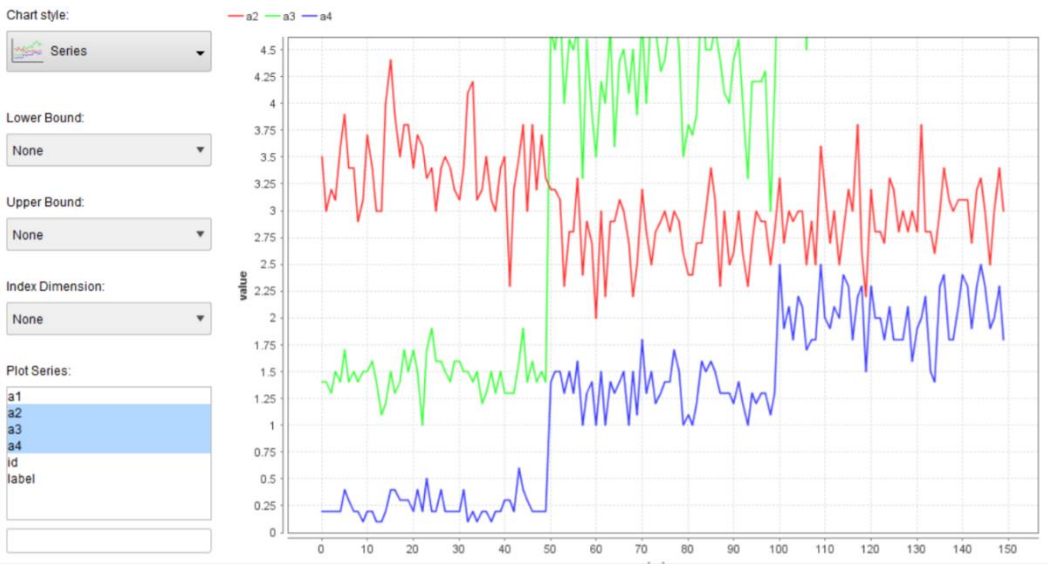


Figure 15: Line Chart

Scatter plots are most useful when the objects in the dataset are static points in time. When plotting time series, line charts are extremely useful.

Histogram

A histogram is a chart that charts the frequency of occurrence of a particular value, similar to a scatter plot. In more detail, this implies that a collection of bins is generated for one of the axes' range values. Consider the iris dataset and the a2, a3, and a4 axis once more. In RapidMiner, go to the charts tab and pick a2, a3, and a4 to plot a histogram. The number of bins can now be set to 40 as illustrated in Figure 16.

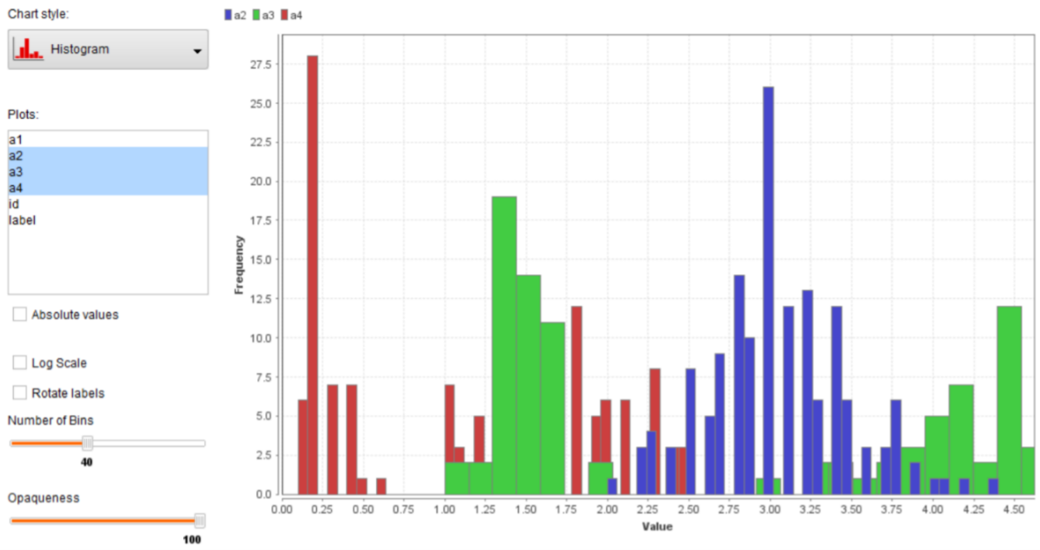


Figure 16:Histogram

Histograms are important to understand how an attribute or a set of attributes is distributed in terms of value.

Pie Chart

A pie chart is typically depicted as a circle divided into sectors, with each sector representing a percentage of a given quantity. Each of the sectors, or slices, is often annotated with a percentage indicating how much of the sum falls into each of the categories. In data analytics, pie charts can be useful for observing the proportion of data points that belong to each of the categories.

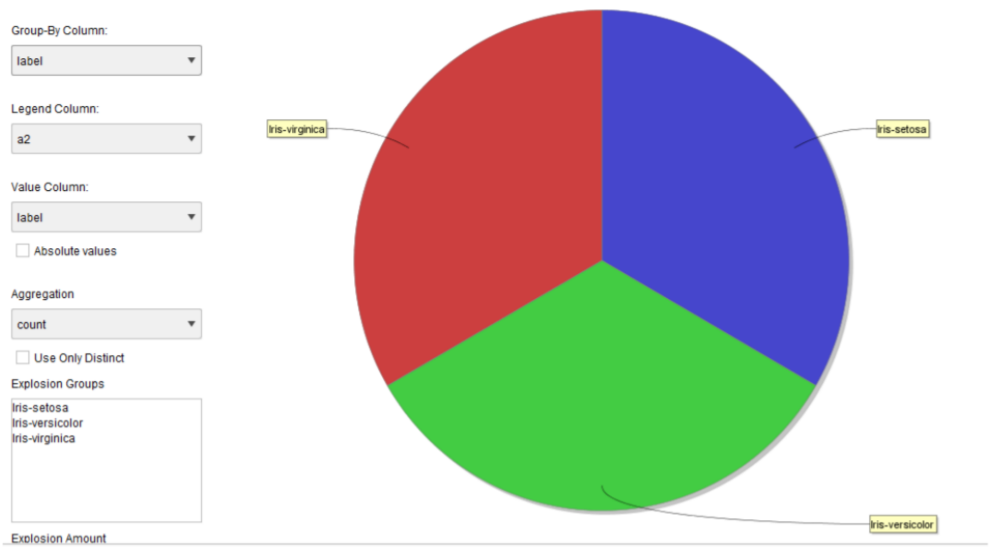


Figure 17:Pie-Chart

To make this map in RapidMiner, go to the Design tab, then to the Charts tab on the top, then to the Pie chart. Pie charts only accept one value as input, and in this case, the labeled column should be considered the input. An example of such an interface is shown in Figure 14. Pie charts, in particular, are a simple way to display how well a dataset is balanced.The Iris dataset is perfectly balanced, as you can see in Figure *17*, with the area equally split between the three classes in the dataset.

Did you Know?To display numerical data, pie graphs are circular graphs divided into sectors or slices.

Box Plots

A box plot is a convenient way to visualize data in terms of its means and quartiles. RapidMiner's box plot is shown in Figure 15. Five main statistics are shown in a box plot: minimum, first quartile, median, third quartile, and maximum. The region of the box is defined by the first and third quartiles, the minimum and maximum are indicated by whiskers, and the median is located within the box plot.White dots are commonly used to represent outliers. Box plots are a useful tool for determining if the dataset's features contain outliers. It's the first step in determining a dataset's quality and whether outlier removal methods are needed to clean it up.

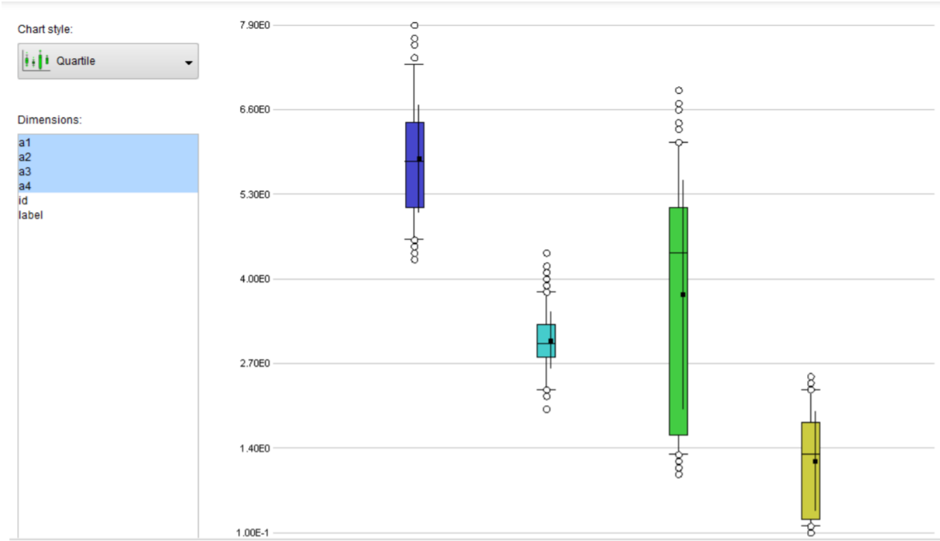
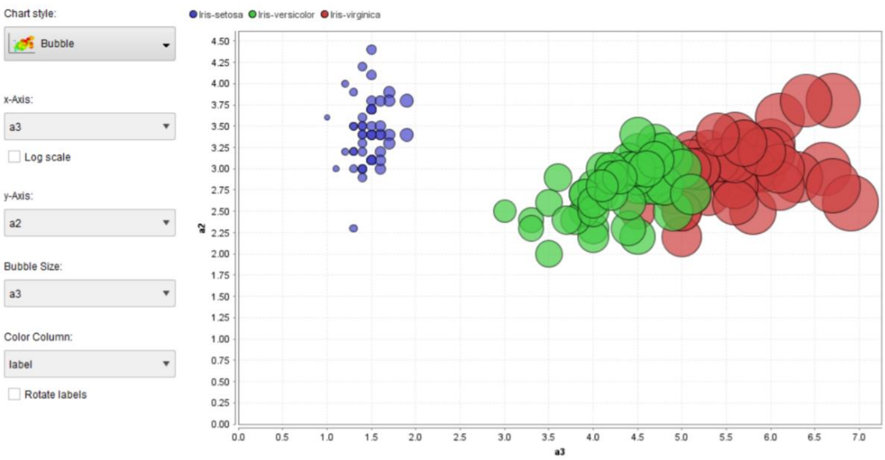


Figure 18:Box Plots

Bubble Charts

Bubble charts can also be modeled with RapidMiner. Bubble charts are a type of two-dimensional diagram that represents three-dimensional data. The ray of a circle represents the third dimension. Using the Iris dataset, Figure 16 shows an example of this diagram in RapidMiner. Bubble charts are important because they provide a two-dimensional representation of the dataset, allowing viewers to maintain a sense of the third dimension based on the size of the circle surrounding the



object.

Figure 19:Bubble Chart of Iris Dataset

Summary

- Store operator stores an IO Object in the data repository.
- The IO Object provided at the input port is delivered through this output port without any modifications.
- The stored object can be used by other processes by using the Retrieve operator.
- The behavior of an Operator can be modified by changing its *parameters*.
- There are two quick and easy ways to store a RapidMiner model in a repository

- There are other different ways of displaying a large number of results, which are also referred to as views within RapidMiner Studio.
- The Retrieve operator loads a RapidMiner Object into the Process.
- Retrieving data this way also provides the metadata of the RapidMiner Object.
- The resulting interface shows you a lot of possibilities in terms of visualization

Keywords

Operators: The elements of a Process, each Operator takes input and creates output, depending on the choice of parameters.

Parameters: Options for configuring the behavior of an Operator.

Help: Displays a help text for the current Operator.

Repository_entry: This parameter is used to specify the location where the input IO Object is to be stored.

Bubble charts: These charts are used for representing three-dimensional data in the form of a two-dimensional diagram.

Self Assessment Questions

1. The Operator that can access stored information in the Repository and load them into the Process.

a) Retrieve Operator
b) Store Operator
c) Rename Operator
d) Filter Examples Operator

2. ____________Contains a large number of operators in order to read data and objects from external formats such as files, databases etc.

a) Export
b) Repository Access
c) Import
d) Evaluation

3.Data, processes, and results are stored in the________

a) Process.
b) Repository.
c) Program
d) RapidMiner Server

4. The *Repository* can be used to store:

a) Data
b) Processes
c) Results
d) All of the above

5. Which of the following ways of getting operator into the Process Panel.

a) Drag-and-drop the Operator
b) Double-click the Operator
c) Right-click the Operator, and choose insert operator from the context menu.
d) All of the above

6. Which of the following format is used for importing data.

a) CSV File

b) Excel Sheet

c) Binary File

d) All

7. You can also load your dataset either from your local system or from a database by clicking on the ___________option.

a) Import Data

b) Export Data

c) Merge Data

d) None

8. RapidMiner Studio can blend structured data with unstructured data and then leverage all the data for _____________analysis.

a) Descriptive

b) Diagnostic

c) Predictive

d) Prescriptive

9. ______________ is an advanced version of RapidMiner Studio that increments the process of building and validating data models.

a) RapidMiner Turbo Prep

b) RapidMiner Auto Model

c) RapidMiner Studio

d) None

10. ______________ RapidMiner Turbo Prep is designed to make the preparation of data much easier.

a) RapidMiner Turbo Prep

b) RapidMiner Auto Model

c) RapidMiner Studio

d) None

11. The _____________shows the individual steps within the analysis process as well as their interconnections.

a) Process View

b) Operator View

c) Repository View

d) None

12. Which of the following is/are used to define an operator?

a) The description of the expected inputs

b) The description of the supplied outputs

c) The action performed by the operator on the inputs, which ultimately leads to the supply of the outputs

d) All of the above

13. Using which of the following option we can delete the selected operator.

a) Pressing the DELETE key

b) Selecting the action "Delete" in the context menu of one of the selected operators

c) Using the menu entry "Edit" – "Delete"

d) All of the Above

14. ________________rearranges all operators of the current process according to the connections and the current execution order.

a) Show and alter execution order
b) Automatic arrangement
c) Automatic size
d) Update projected metadata

15. ____________is not dedicated to pre-defined descriptions but rather to your own comments on individual steps of the process.

a) Meta-Data View
b) Help View
c) Comment View
d) Problems View

Answer for Self Assessment

1.	A	2.	C	3.	B	4.	D	5.	D
6.	D	7.	A	8.	C	9.	B	10.	A
11.	A	12.	D	13.	D	14.	B	15.	C

Review Questions

Q1) How to connect with the data using Rapidminer Studio. What are the different file formats supported by Rapidminer?

Q2) Write down the different ways to store a RapidMiner model in a repository.

Q3) Create your won dataset and write down the steps on how you can import and store the data created by you using the store and import operator.

Q4) With an appropriate example explain the different ways of importing the data in rapidminer studio.

Q4) When and how to how to add a 'label' attribute to a dataset? Explain with example?

Q5) Create your repository and load data into it. Process the data and represent the results using three different charts available in rapidminer.

Q6) Write down the step-by-step procedure to display the data using a line chart.

Further Readings

Kotu, V., & Deshpande, B. (2014). Predictive analytics and data mining: concepts and practice with rapidminer. Morgan Kaufmann.

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